

Name : Vũ Đức Tùng

IRN: 2331300132

Course : CSE 203

Lab 1

Problem 1:

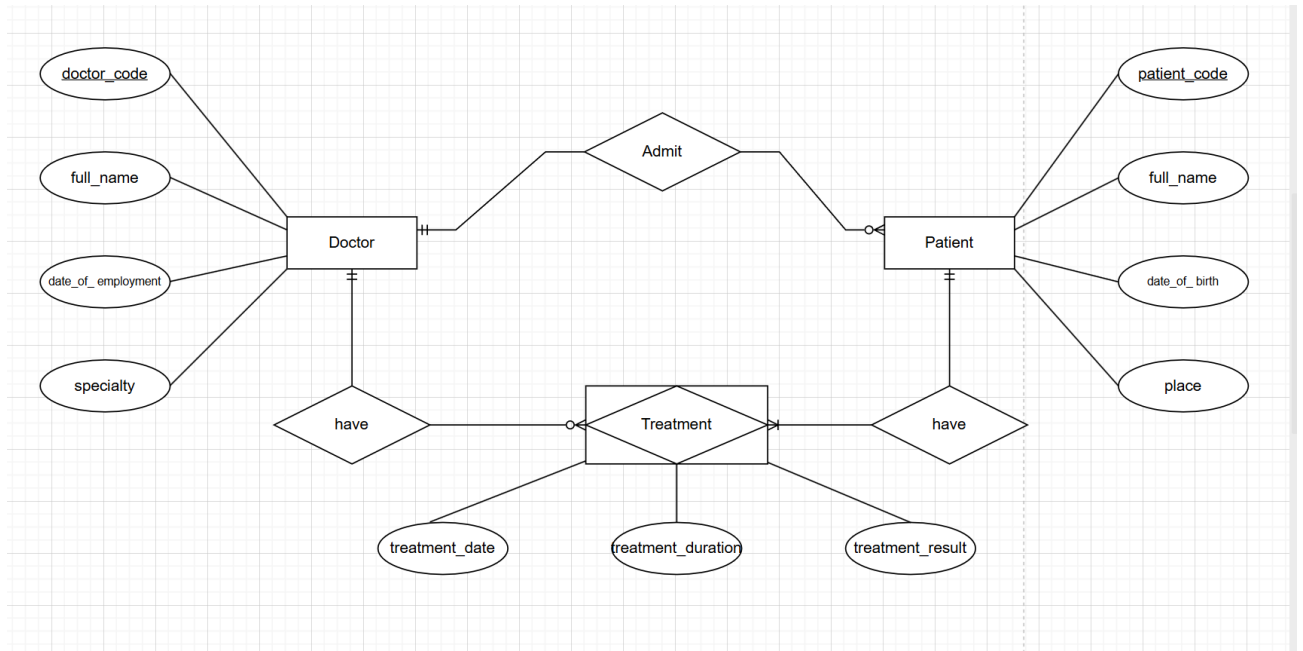
1. Entities and attributes:

- a. Doctor (doctor_code, full_name, date_of_employment, specialty)
- b. Patient (patient_code, full_name, date_of_birth, place)
- c. Record (treatment_date , treatment_duration, treatment_result)

2. Relation:

- a. Doctor – Patient: Admit (1 : n)
- b. Doctor – Patient: Treat (n : n)
- c. Doctor – Record: (1 : n)
- d. Patient – Record : (1 : n)

3. ERD



Problem 2:

1. Entities and attributes:
 - a. faculty (faculty_code, faculty_name)
 - b. department (department_code , department_name)
 - c. class (class_code, class_name, numbers_of_students)
 - d. student (student_id, full_name, date_of_birth , gender , address)
 - e. subject (subject_code , subject_name, number_of_credits)
 - f. library_card (card_number, issue_date, expiration_date)
 - g. exam_result (exam_score , graduation_score)
2. Relation:
 - a. faculty – department : (1 : n)
 - b. department – class : (1: n)
 - c. student – class : (n :1)
 - d. student - subject : (m: n)
 - e. subject – exam_result : (1 : 1)
 - f. student – library_card (1 : 1)
3. Weak entity:
 - a. Weak entity : library_card
 - b. Associative entity : exam results
4. ERD



Problem 3

1. Entity and attribute

- room (room_code, room_name, floor , daily_rate, weekly_rate)
- floor (floor_code, floor_name)
- guest (guest_identity, full_name, phone_number)
- invoice(invoice_code, guest_information, room_used, arrival_date, departure_date, room_charge, total_service_charge, final_total)
- service (service_code , service_name , unit_price)
- service_record (invoice_code , service_code , quantity)

2. Relation

- floor – room : have (1:n)
- guest - invoice : have (1 : n)
- room - invoice : have (1:n)
- guest – service : (1:n)
- service – service_record : have (1 : n)
- invoice – service_record : use (1 : n)

3. Weak entity

- service_record.

4. ERD

